1. (例 6.3.2)

$$
\begin{aligned}
I &= \int \frac{x^3+1}{x^4-3x^3+3x^2-x}\mathrm{d}x \\
&= \int \frac{x^3+1}{x(x-1)^3}\mathrm{d}x \\
&= \int \left(\frac{A}{x}+\frac{B}{x-1}+\frac{C}{(x-1)^2}+\frac{D}{(x-1)^3}\right)\mathrm{d}x.
\end{aligned}
$$

其中, $A(x-1)^3+Bx(x-1)^2+Cx(x-1)+Dx=x^3+1$, 即

$$
\begin{cases}
-A=1, \\
3A+B-C+D=0, \\
-3A-2B+C=0, \\
A+B=1.
\end{cases}
\Rightarrow
\begin{cases}
A=-1, \\
B=2, \\
C=1, \\
D=2.
\end{cases}
$$

所以

$$
\begin{aligned}
I &= -\int \frac{\mathrm{d}x}{x}+2\int \frac{\mathrm{d}x}{x-1}+\int \frac{\mathrm{d}x}{(x-1)^2}+2\int \frac{\mathrm{d}x}{(x-1)^3} \\
&= -\ln|x|+2\ln|x-1|-\frac{1}{x-1}-\frac{1}{(x-1)^2}+C.
\end{aligned}
$$

2. (例 6.3.6)

$$
\begin{aligned}
I &= \int \frac{\cos^2 x}{(a^2\sin^2 x+b^2\cos^2 x)^2}\mathrm{d}x \\
&= \int \frac{\cos^4 x}{(a^2\sin^2 x+b^2\cos^2 x)^2\cos^2 x}\mathrm{d}x \\
&= \int \frac{1}{(a^2\tan^2 x+b^2)^2}\mathrm{d}\tan x \\
&= \frac{1}{a^4}\int \frac{1}{\left(t^2+k^2\right)^2}\mathrm{d}t. \quad (t=\tan x,\ k=b^2/a^2)
\end{aligned}
$$

回忆有, 对于以下积分:

$$
I_n = \int \frac{1}{(x^2+a^2)^n}\mathrm{d}x,
$$

给出递推公式

$$
\begin{aligned}
I_n &= \frac{x}{(x^2+a^2)^n}-\int x\mathrm{d}\frac{1}{(x^2+a^2)^n} \\
&= \frac{x}{(x^2+a^2)^n}+2n\int \frac{x^2}{(x^2+a^2)^{n+1}} \\
&= \frac{x}{(x^2+a^2)^n}+2n(I_n-a^2I_{n+1}),
\end{aligned}
$$

即

$$
I_{n+1} = \frac{1}{2na^2}\left(\frac{x}{(x^2+a^2)^n}+(2n-1)I_n\right).
$$

边界有

$$I_1 = \frac{1}{a}\int \frac{1}{\frac{x^2}{a^2}+1}\mathrm{d}\frac{x}{a} = \frac{1}{a}\arctan\frac{x}{a} + C.$$

回到原题, 代入公式有

$$\begin{aligned} I_1 &= \frac{1}{k}\arctan\frac{x}{k} + C, \\ I_2 &= \frac{1}{2k^2}\left(\frac{x}{x^2+k^2} + \frac{1}{k}\arctan\frac{x}{k}\right) + C. \end{aligned}$$

因此最终答案

$$\begin{aligned} I &= \frac{1}{a^4}\cdot\frac{a^2}{2b^2}\left(\frac{a^2t}{a^2t^2+b^2} + \frac{a}{b}\arctan\frac{at}{b}\right) + C \\ &= \frac{1}{2a^2b^2}\left(\frac{a^2\tan x}{a^2\tan^2 x + b^2} + \frac{a}{b}\arctan\left(\frac{a}{b}\tan x\right)\right) + C. \end{aligned}$$

3. (例 6.3.8)

$$\begin{aligned} I &= \int \frac{1-\sqrt{x+1}}{1+\sqrt[3]{x+1}}\mathrm{d}x \\ &= 6\int \frac{-t^8+t^5}{t^2+1}\mathrm{d}t \\ &= 6\int \left(-t^6+t^4+t^3-t^2-t+1+\frac{t-1}{t^2+1}\right)\mathrm{d}t \\ &= 6\int \left(-t^6+t^4+t^3-t^2-t+1\right)\mathrm{d}t + 3\int \frac{\mathrm{d}t^2}{t^2+1} - 6\int \frac{\mathrm{d}t}{t^2+1} \\ &= -\frac{6}{7}(x+1)^{\frac{7}{6}} + \frac{6}{5}(x+1)^{\frac{5}{6}} + \frac{3}{2}(x+1)^{\frac{2}{3}} - 2(x+1)^{\frac{1}{2}} \\ &\quad - 3(x+1)^{\frac{1}{3}} + 6(x+1)^{\frac{1}{6}} + 3\ln\left|(x+1)^{\frac{1}{3}}+1\right| - 6\arctan(x+1)^{\frac{1}{6}}. \end{aligned}$$

4. (习题六 6)

(4)

$$\begin{aligned} I &= \int \frac{2x^2+2x+13}{(x-2)(1+x^2)^2}\mathrm{d}x \\ &= \int \left(\frac{A}{x-2} + \frac{Bx+C}{1+x^2} + \frac{Dx+E}{(1+x^2)^2}\right)\mathrm{d}x. \end{aligned}$$

其中,

$$A(x^2+1)^2 + (Bx+C)(x-2)(x^2+1) + (Dx+E)(x-2) = 2x^2+2x+13,$$

即

$$\begin{cases} A+B=0, \\ -2B+C=0, \\ 2A+B-2C+D=2, \\ -2B+C-2D+E=2, \\ A-2C-2E=13 \end{cases} \Rightarrow \begin{cases} A=1, \\ B=-1, \\ C=-2, \\ D=-3, \\ E=-4. \end{cases}$$

因此

$$
\begin{aligned}
I &= \int \left(\frac{1}{x-2} - \frac{x+2}{x^2+1} - \frac{3x+4}{(x^2+1)^2} \right) \mathrm{d}x \\
&= \int \frac{\mathrm{d}x}{x-2} - \frac{1}{2} \int \frac{\mathrm{d}x^2}{x^2+1} - 2 \int \frac{\mathrm{d}x}{x^2+1} - \frac{3}{2} \int \frac{\mathrm{d}x^2}{(x^2+1)^2} - 4 \int \frac{\mathrm{d}x}{(x^2+1)^2} \\
&= \ln|x-2| - \frac{1}{2} \ln(x^2+1) - 2\arctan x + \frac{3}{4(x^2+1)} - \frac{1}{2} \left(\frac{x}{x^2+1} + \arctan x \right) + C \\
&= \ln \frac{|x-2|}{\sqrt{x^2+1}} - \frac{5}{2} \arctan x + \frac{3-2x}{4(x^2+1)} + C.
\end{aligned}
$$

5. (习题六 7)

(10)

$$
\begin{aligned}
I &= \int \frac{\sin x}{\sin x - \cos x + 2} \mathrm{d}x \\
&= \int \frac{\frac{2t}{1+t^2}}{\frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} + 2} \frac{2}{1+t^2} \mathrm{d}t \\
&= 4 \int \frac{t \mathrm{d}t}{(1+t^2)(2t - (1-t^2) + 2(1+t^2))} \\
&= 4 \int \frac{t \mathrm{d}t}{(1+t^2)(3t^2+2t+1)} \\
&= 4 \int \left(\frac{At+B}{1+t^2} + \frac{Ct+D}{3t^2+2t+1} \right) \mathrm{d}t.
\end{aligned}
$$

其中

$$
\begin{gathered}
(At+B)(3t^2+2t+1) + (Ct+D)(1+t^2) = t \\
\Rightarrow \begin{cases} 3A + C = 0, \\ 2A + 3B + D = 0, \\ A + 2B + C = 1, \\ B + D = 0 \end{cases} \Rightarrow \begin{cases} A = -\frac{1}{4}, \\ B = \frac{1}{4}, \\ C = \frac{3}{4}, \\ D = -\frac{1}{4}. \end{cases}
\end{gathered}
$$

代回得

$$
\begin{aligned}
I &= \int \frac{1-t}{1+t^2} \mathrm{d}t + \int \frac{3t-1}{3t^2+2t+1} \mathrm{d}t \\
&= \arctan t - \frac{1}{2} \ln(1+t^2) + \int \frac{t - \frac{1}{3}}{\left(t + \frac{1}{3}\right)^2 + \frac{2}{9}} \mathrm{d}t \\
&= \arctan t - \frac{1}{2} \ln(1+t^2) + \frac{1}{2} \int \frac{\mathrm{d}\left(t + \frac{1}{3}\right)^2}{\left(t + \frac{1}{3}\right)^2 + \frac{2}{9}} - 3 \int \frac{\mathrm{d}t}{\left(\frac{3t+1}{\sqrt{2}}\right)^2 + 1} \\
&= \arctan t - \frac{1}{2} \ln(1+t^2) + \frac{1}{2} \ln \left| t^2 + \frac{2}{3} t + \frac{1}{3} \right| - \sqrt{2} \int \frac{\mathrm{d}\left(\frac{3t+1}{\sqrt{2}}\right)}{\left(\frac{3t+1}{\sqrt{2}}\right)^2 + 1} \\
&= \frac{x}{2} - \frac{1}{2} \ln \left(1 + \tan^2 \frac{x}{2}\right) + \frac{1}{2} \ln \left| \tan^2 \frac{x}{2} + \frac{2}{3} \tan \frac{x}{2} + \frac{1}{3} \right| - \sqrt{2} \arctan \frac{3 \tan \frac{x}{2} + 1}{\sqrt{2}} + C.
\end{aligned}
$$

6. (习题六 8)

(2)

$$
\begin{aligned}
\int \frac{\mathrm{d}x}{1+\sqrt{x}+\sqrt{x+1}} &= \int \frac{\sqrt{x}+1-\sqrt{x+1}}{2\sqrt{x}}\mathrm{d}x \\
&= \frac{x}{2}+\sqrt{x}-\frac{1}{2}\int \frac{\sqrt{x+1}}{\sqrt{x}}\mathrm{d}x \\
&= \frac{x}{2}+\sqrt{x}-\frac{1}{2}\sqrt{x(x+1)}+\frac{1}{2}\int\left(\frac{\sqrt{x}}{2\sqrt{x+1}}-\frac{\sqrt{x+1}}{2\sqrt{x}}\right)\mathrm{d}x \\
&= \frac{x}{2}+\sqrt{x}-\frac{1}{2}\sqrt{x(x+1)}-\frac{1}{4}\int \frac{\mathrm{d}x}{\sqrt{x(x+1)}} \\
&= \frac{x}{2}+\sqrt{x}-\frac{1}{2}\sqrt{x(x+1)}-\frac{1}{2}\frac{\mathrm{d}(\sqrt{x}+\sqrt{x+1})}{\sqrt{x}+\sqrt{x+1}} \\
&= \frac{x}{2}+\sqrt{x}-\frac{1}{2}\sqrt{x(x+1)}-\frac{1}{2}\ln(\sqrt{x}+\sqrt{x+1})+C.
\end{aligned}
$$